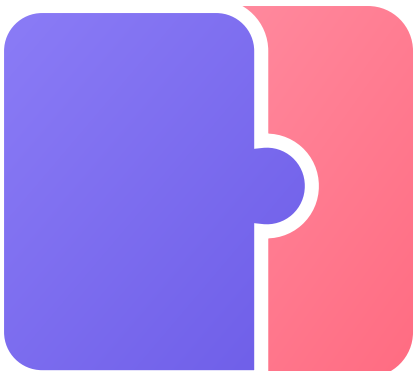
TeachReach

A social network connecting experts with those who need their knowledge

Logo



What We’ll Cover

Problem & Idea

Key Features

Architecture

Problem & Idea

The Problem

Those who offer knowledge

- Struggle to find people who actually need what they know
- Advertising is scattered across dozens of platforms and groups
- No clear, structured way to showcase what they offer

Those who need knowledge

- Don’t know who to turn to
- Search chaotically, with no filtering by area of interest
- Waste time before finding the right person

Solution: TeachReach

A social network that automatically connects people who **provide** services/tutoring with people who are **looking** for them – based on interest tags.

1

Sign-up & Profile

Cognito + interest tags

2

Smart Classification

Bedrock identifies the service topic

3

Matching

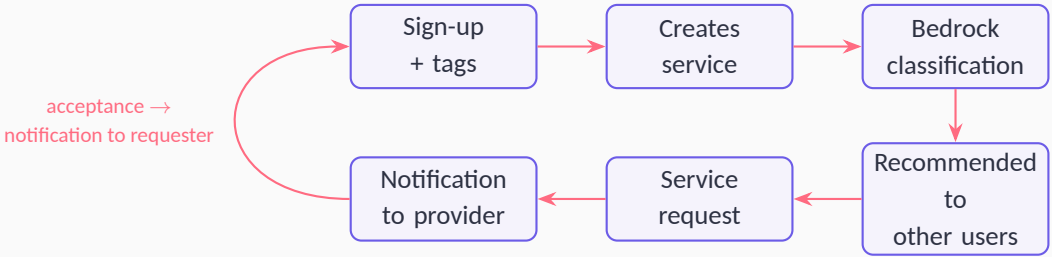
Request → acceptance → contact

Key Features

5 Key Features

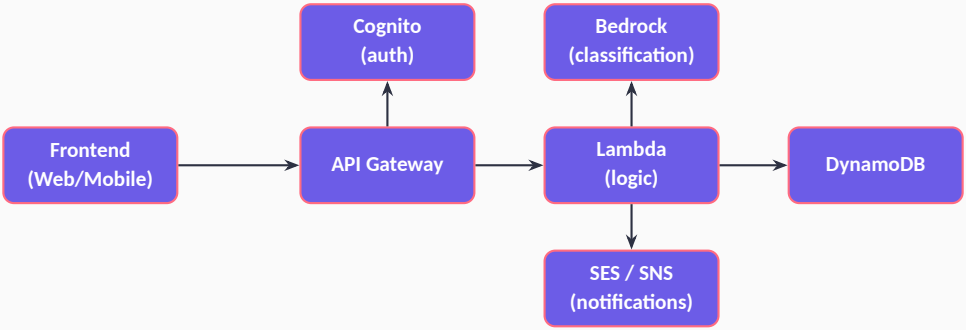
1. **Sign-up and login** – AWS Cognito, with email notification of registration status
2. **Interest profile** – the user selects tag(s) during registration
3. **Service creation** – description + tag, chosen manually by the user *or* automatically classified by AWS Bedrock
4. **Personalized recommendations** – posts are filtered according to the user's interest tags
5. **Request system** – service request → notification to provider → acceptance → notification to requester

User Flow


Sign-up
+ tags
Creates
service
Bedrock
classification
Recommended
to
other users
Service
request
Notification
to provider
acceptance →
notification to requester

Architecture

AWS Architecture



AWS Services Used

Service	Role
Cognito	Sign-up, login, user management
SES	Email notifications for registration status and requests
Lambda	Business logic (serverless)
API Gateway	REST API layer
DynamoDB	Database (users, services, requests)
Bedrock	Classifies service description into a tag
SNS	Push notifications for requests

Data Model (DynamoDB – single table)

Entity	PK / SK	GSI
User	`USER#id`	`TAG#interest`
Service	`SERVICE#id`	`TAG#category`
Request	`REQUEST#id`	`USER#id (status)`

- Single table, GSI by tag enables fast recommendations
- Request status: `pending` → `accepted` / `declined`

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TeachReach

connecting knowledge with those who need it